Bioinformatics analysis of six *Candida parapsilosis* genome assemblies obtained using whole-genome optical maps and Illumina short-read data

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INTRODUCTION AND OBJECTIVES

The use of Next Generation Sequencing (NGS) technologies has laid the basis for the implementation of many different algorithms for both *de novo* and reference-guided genomes reconstruction.

De novo genome assembly, using solely short reads, lacks of a good resolving power due to repetitive DNA sequences present in genomes (and reads) and high sequencing error rates that frequently occurr in GC- or AT-rich regions, leading to incomplete and fragmented contigs/scaffolds containing misassembled regions and incorrected bases [4]. On the contrary, the use of a reference genome with potentially misassembled chromosomal regions and/or with the troubling presence of structural rearrangements with respect to the test sample, can lead to wrong interpretations [14]. Moreover, highly divergent DNA regions may not be reconstructed. Nevertheless, our requirement of maintaining the genetic differences of each of the six fungal strains examined, relative to the target genome, prompted us to use a hybrid approach between the *de novo* genome assembly and the reference-guided one, taking the best from both.

Here, we developed a new pipeline (Fig.3) for a hybrid *de novo*/reference-guided genome assembly by combining both whole-genome optical mapping (WGM) technology [11] and NGS for the study of six *Candida parapsilosis* strains (WGCP1-6), collected from different sources (hand, air, sink, central venous catheter (CVC), blood) during 2016-2017. *C. parapsilosis*, is a clonal pathogen whose relevance relies on its ability to form tenacious biofilms on CVCs and other medically implanted devices [3, 10].

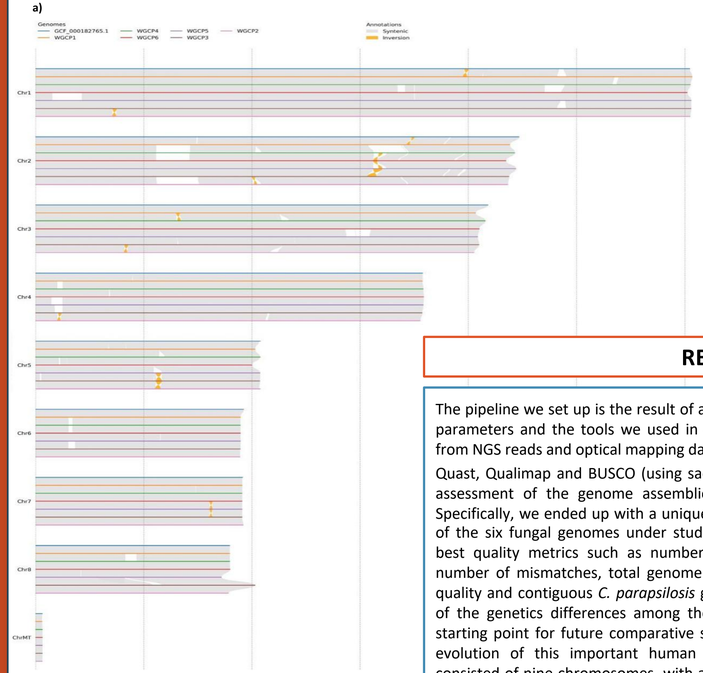
MATERIAL AND METHODS

From all the six fungal isolates, DNA was extracted, and then genomic libraries were sequenced on Illumina NovaSeq platform using a 2x150bp sequencing approach. At the same time, the protoplasts were isolated to produce high molecular weight DNA necessary for digestion with BspQI enzyme in order to construct ordered, genome-wide, high-resolution restriction maps by optical mapping.

Illumina short paired-end reads were checked for their quality, clipped to get rid of adaptors, trimmed for low quality bases at 3' and 5' and filtered out for contaminants and low-quality regions by using the fastp (v0.23.1) [5] algorithm. Clean reads were assembled by three different NGS genome assemblers: SPAdes (v.3.15) [17], MEGAHIT (v.1.2.9) [7] and SOAPdenovo2 (v.2.04) [16] using different kmer sizes (25 to 125, step size 10) to reconstruct the genome assembly at scaffolds level. Then, we selected two best assemblies for each strain that were merged together by Quickmerge tool in order to get complementary information from different assemblers and overcome the issue of incompleteness and fragmentation. We performed a benchmarking analysis on the parameters to be setted for the assembly reconciliation performed by Novo&Stitch [8], this tool uses indeed optical maps and the merged *de novo* genome assembly produced with short reads and returns a very accurate and contiguous genome assembly derived from the combination of data from the two complementary technologies.

Draft assemblies of the six *C. parapsilosis* strains were used to infer a distance-based phylogenetic tree (Fig.2) with branch supports using JolyTree software (v.1.1b) [1], which was subsequently used to identify a representative isolate assembly that better summarizes the overall genetic variability within the species. Primary assemblies of this representative strain (WGCP4), resulting from optical and NGS genome reconstructions, were merged together using QuickMerge (v.3), and the resulting hybrid assembly was scaffolded with SSPACE (v.3.0) [15], a reference-free tool that uses just paired-end reads.

A second run of scaffolding was executed on WGCP4, this time with NtJoin algorithm (v.1.1.1) [9] that takes the reference ASM18276v2 of *C. parapsilosis,* deposited in the NCBI, and uses its information to scaffold the target assembly using minimizer graphs to yield a mapping between the input assemblies. The same operation was carried out with the remaining five isolates, using the reconstructed WGCP4 assembly itself as reference for the scaffolding. Non-nuclear contigs were assembled using MITObim for the reconstruction of mitochondrial genomes from NGS reads and the ASM18276v2 reference mitochondrial chromosome was used as a bait. Finally, nuclear and non-nuclear sequences were further polished by filling, with GapFIller (v.1.10) [6], the large amount of gaps due to the long distance between BspQI restriction enzyme sites along the genomes that were used to produce the optical maps. Technical metrics were computed at each step of our pipeline in order to evaluate the quality of each assembly, using tools such as: Busco (v.5.4.3) [2], which estimates the completeness and redundancy of genome assemblies by providing biologically meaningful metrics based on expected gene content; Qualimap (v.2.2.2) [12], which takes sequence alignment data and provides quality-control data; and Quast (v.5.2.0) [13] which evaluates assemblies and produces many reports, summary tables and plots.



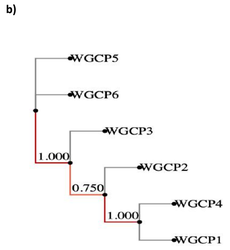


Fig. 1 a) Genomic differences computed with SyRi [18] between WGCP1-6 genomes using reconstructed assemblies; **b)** Phylogenetic tree produced with JolyTree.

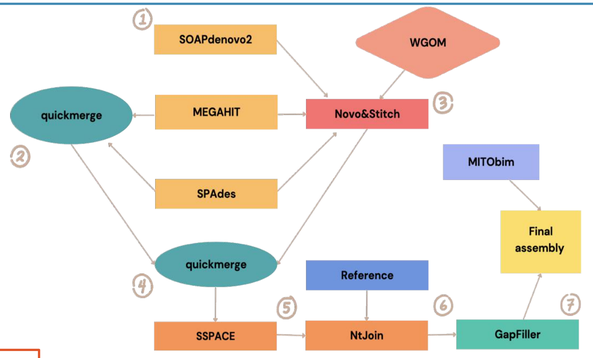


Fig. 3 Overview of the principal steps involved in the developed pipeline.

QUALIMAP	WGCP1	WGCP2	WGCP3	WGCP4	WGCP5	WGCP6
number of bases (bp)	12,897,454	12,882,277	13,043,336	12,985,286	12,920,649	12,867,357
number of contigs	9	9	9	9	9	9
number of reads	13085167	12284976	12266065	10958058	11128277	11547894
number of mapped reads	94.72%	95.54%	96.12%	96.93%	95.8%	96.8%
mean mapping quality	57,3632	57,6416	56,8987	58,2625	58,3136	58,8408
QUAST	**WGCP1**	**WGCP2**	**WGCP3**	**WGCP4**	**WGCP5**	**WGCP6**
Largest contig	3034869	3025418	3029999	3026431	3029950	3012190
N50	2034685	2026981	2049707	2079053	2043907	2050912
N90	947582	945172	954645	946945	947745	947508
L50	3	3	3	3	3	3
BUSCO	**WGCP1**	**WGCP2**	**WGCP3**	**WGCP4**	**WGCP5**	**WGCP6**
BUSCO groups searched	2137	2137	2137	2137	2137	2137
Complete %	93,80%	94,00%	95,90%	96,20%	95,40%	95,20%
Complete and single-copy %	93,40%	93,60%	95,50%	95,70%	95,30%	95,10%
Complete and duplicated %	0,40%	0,04%	0,40%	0,50%	0,10%	0,10%
Fragmented %	0,60%	0,50%	0,50%	0,50%	0,50%	0,70%
Missing %	5,60%	5,50%	3,60%	3,30%	4,10%	4,10%

Table 1 Metrics of the six C. *parapsilosis* genomes.

RESULTS

The pipeline we set up is the result of a benchmarking analysis operated on both the parameters and the tools we used in order to produce a hybrid assembly starting from NGS reads and optical mapping data.

Quast, Qualimap and BUSCO (using saccharomycetes_odb10) were used for quality assessment of the genome assemblies produced at each step of our pipeline. Specifically, we ended up with a unique workflow that completely reconstructs each of the six fungal genomes under study producing genome assemblies with overall best quality metrics such as number of contigs/scaffolds, N50 and N90 values, number of mismatches, total genome length and N-values. The resulting six high-quality and contiguous *C. parapsilosis* genome assemblies give a complete overview of the genetics differences among the strains investigated, and provide a robust starting point for future comparative studies to better understand the biology and evolution of this important human fungal pathogen. The final six assemblies consisted of nine chromosomes, with a total length spanning from ~12.86 to ~13.05 Mbp, N50 values between 2,026,916 and 2,079,028, and a BUSCO genome completeness score ranging from 93.8 to 96.2% (Table 1).

DISCUSSION AND CONCLUSION

Based on a deep benchmarking analysis of the best parameters and softwares for de novo genome assembly starting from short reads and optical mapping data, we obtained six high-quality genome assemblies of *C. parapsilosis* at chromosome level.

We set a unique pipeline based on the result obtained on a single fungal strain (WGCP4); then we applied the same workflow over the other 5 strains, ending up with overall good results as well.

The resulting complete and accurate genomes ultimately prove the suitability of our pipeline for genome assembly from optical mapping and NGS data.

It is indeed very convenient to set in advance the best parameters and the best set of operations to execute in order to produce an accurate genome assembly at chromosome level, and the reproducibility of our result over each of the 6 strains definitely proves the suitability of our approach.

REFERENCES

[1] Criscuolo, A. A Fast Alignment-Free Bioinformatics Procedure to Infer Accurate Distance-Based Phylogenetic Trees from Genome Assemblies. *RIO*, 2019, *5*, e36178.

[2] Simão, F. A.; Waterhouse, R. M.; Ioannidis, P.; Kriventseva, E. V.; Zdobnov, E. M. BUSCO: Assessing Genome Assembly and Annotation Completeness with Single-Copy Orthologs. *Bioinformatics*, 2015, *31* (19), 3210–3212.

[3] Tóth, R.; Nosek, J.; Mora-Montes, H. M.; Gabaldon, T.; Bliss, J. M.; Nosanchuk, J. D.; Turner, S. A.; Butler, G.; Vágvölgyi, C.; Gácser, A. Candida Parapsilosis: From Genes to the Bedside. *Clin Microbiol Rev*, 2019, *32* (2), e00111-18.

[4] Chakraborty, M.; Baldwin-Brown, J. G.; Long, A. D.; Emerson, J. J. Contiguous and Accurate *de Novo* Assembly of Metazoan Genomes with Modest Long Read Coverage. *Nucleic Acids Res*, 2016, gkw654.

[5] Chen, S.; Zhou, Y.; Chen, Y.; Gu, J. Fastp: An Ultra-Fast All-in-One FASTQ Preprocessor. *Bioinformatics*, 2018, *34* (17), i884–i890.

[6] Nadalin, F.; Vezzi, F.; Policriti, A. GapFiller: A de Novo Assembly Approach to Fill the Gap within Paired Reads. *BMC Bioinformatics*, 2012, *13* (S14), S8.

[7] Li, D.; Liu, C.-M.; Luo, R.; Sadakane, K.; Lam, T.-W. MEGAHIT: An Ultra-Fast Single-Node Solution for Large and Complex Metagenomics Assembly via Succinct de Bruijn Graph. *Bioinformatics*, 2015, *31* (10), 1674–1676.

[8] Pan, W.; Wanamaker, S. I.; Ah-Fong, A. M. V.; Judelson, H. S.; Lonardi, S. Novo&Stitch: Accurate Reconciliation of Genome Assemblies via Optical Maps. *Bioinformatics*, 2018, *34* (13), i43–i51.

[9] Coombe, L.; Nikolić, V.; Chu, J.; Birol, I.; Warren, R. L. NTJoin: Fast and Lightweight Assembly-Guided Scaffolding Using Minimizer Graphs. *Bioinformatics*, 2020, *36* (12), 3885–3887.

[10] Kocsubé, S.; Tóth, M.; Vágvölgyi, C.; Dóczi, I.; Pesti, M.; Pócsi, I.; Szabó, J.; Varga, J. Occurrence and Genetic Variability of Candida Parapsilosis Sensu Lato in Hungary. *Journal of Medical Microbiology*, 2007, *56* (2), 190–195.

[11] Leinonen, M.; Salmela, L. Optical Map Guided Genome Assembly. *BMC Bioinformatics*, 2020, *21* (1), 285.

[12] García-Alcalde, F.; Okonechnikov, K.; Carbonell, J.; Cruz, L. M.; Götz, S.; Tarazona, S.; Dopazo, J.; Meyer, T. F.; Conesa, A. Qualimap: Evaluating next-Generation Sequencing Alignment Data. *Bioinformatics*, 2012, *28* (20), 2678–2679.

[13] Gurevich, A.; Saveliev, V.; Vyahhi, N.; Tesler, G. QUAST: Quality Assessment Tool for Genome Assemblies. *Bioinformatics*, 2013, *29* (8), 1072–1075.

[14] Volarić, M.; Despot-Slade, E.; Veseljak, D.; Meštrović, N.; Mravinac, B. Reference-Guided De Novo Genome Assembly of the Flour Beetle Tribolium Freemani. *Int J Mol Sci*, 2022, *23* (11), 5869.

[15] Boetzer, M.; Henkel, C. V.; Jansen, H. J.; Butler, D.; Pirovano, W. Scaffolding Pre-Assembled Contigs Using SSPACE. *Bioinformatics*, 2011, *27* (4), 578–579.

[16] Luo, R.; Liu, B.; Xie, Y.; Li, Z.; Huang, W.; Yuan, J.; He, G.; Chen, Y.; Pan, Q.; Liu, Y.; et al. SOAPdenovo2: An Empirically Improved Memory-Efficient Short-Read de Novo Assembler. *GigaSci*, 2012, *1* (1), 18.

[17] Bankevich, A.; Nurk, S.; Antipov, D.; Gurevich, A. A.; Dvorkin, M.; Kulikov, A. S.; Lesin, V. M.; Nikolenko, S. I.; Pham, S.; Prjibelski, A. D.; et al. SPAdes: A New Genome Assembly Algorithm and Its Applications to Single-Cell Sequencing. *Journal of Computational Biology*, 2012, *19* (5), 455–477.

[18] Goel, M.; Sun, H.; Jiao, W.-B.; Schneeberger, K. SyRI: Finding Genomic Rearrangements and Local Sequence Differences from Whole-Genome Assemblies. *Genome Biol*, 2019, *20* (1), 277.